

SHIP WORK BREAKDOWN STRUCTURE TITLE

Group 0-99 General Guidance and Administration

000	General Guidance and Administration
010	Combat Capabilities (Offensive and Defensive)
011	Air Weapons Vs. Air Targets
012	Air Weapons Vs. Surface Targets
013	Air Weapons Vs. Underwater Targets
014	Surface Weapons Vs. Air Targets
015	Surface Weapons Vs. Surface Targets
016	Surface Weapons Vs. Underwater Targets
017	Underwater Weapons Vs. Surface Targets
018	Underwater Weapons Vs. Underwater Targets
020	Strategic and Special Capabilities
021	Surface Based Deterrents
022	Underwater Based Deterrents
023	Amphibious Warfare
024	Mine and Mine Countermeasure Warfare
025	Inshore Warfare
030	Tactical and Strategic Oper. Support Capabilities
031	Command/Control/Communications
032	Surveillance/Reconnaissance/Intelligence
033	Electronic Warfare and Nuc/Bio/Chemical Defense
034	Logistics/Sealift
035	Other Support
040	Ship System Management
041	Project Management
042	General Administrative Requirements
043	Life Cycle Costing
044	Ship Operation
050	Ship System Performance
051	Ship System Performance Concepts
052	Ship Subsystem Performance Concepts
053	Selected Concepts
054	Component Development
060	Subsystem Characteristics (Interfaces and CNO Cont.)
061	Hull Structure
062	Propulsion Plant
063	Electric Plant
064	Command and Surveillance
065	Auxiliary Systems
066	Outfitting
067	Weapons
068	Integration and Engineering

SHIP WORK BREAKDOWN STRUCTURE TITLE

069	Ship Assembly
070	General Requirements for Design and Construction
071	Access
072	Shock
073	Noise and Vibration
074	Casting, Welding, Riveting, Allied Processes (General)
075	Threaded Fasteners Standards
076	Reliability ad Maintainability
077	Safety
078	Materials
079	Seaworthiness
080	Integrated Logistic Support Requirements
081	Maintenance
082	Support and Test Equipment
083	supply support
084	Transportation and Handling
085	Engineering Drawings
086	Technical Manuals and Other Data
087	Facilities
088	Personnel and Training
089	Training Equipment
090	Quality Assurance Requirements
091	Ship Inspections
092	Ship Tests
093	Combat Systems Checkout
094	Regular Ship Trials
095	Whole Ship Testing
096	Weight Control
097	Inclining Experiment and Trim Dive
098	Models and Mockups
099	Photographs

Group 1 Hull Structure

100	Hull Structure, General
101	General Arrangement-Structural Drawings
110	Shell and Support Structure
111	Shell Plating, Surf. Ship and Submarine Press. Hull
112	Shell Plating, Submarine Non-pressure Hull
113	Inner Bottom
114	Shell Appendages
115	Stanchions

SHIP WORK BREAKDOWN STRUCTURE TITLE

116	Longit. Framing, Surf. Ship and Submarine Press. Hull
117	Transv. Framing, Surf. Ship and Submarine Press. Hull
118	Longit. and Transv. Submarine Non-press. Hull Framing
119	Lift System Flexible Skirts and Seals
120	Hull Structural Bulkheads
121	Longitudinal Structural Bulkheads
122	Transverse Structural Bulkheads
123	Trunks and Enclosures
124	Bulkheads in Torpedo Protection System
125	Submarine Hard Tanks
126	Submarine Soft Tanks
130	Hull Decks
131	Main Deck
132	2nd Deck
133	3rd Deck
134	4th Deck
135	5th Deck and Decks Below
136	01 Hull Deck (Forecastle and Poop Decks)
137	02 Hull Deck
138	03 Hull Deck
139	04 Hull Deck and Hull Decks Above
140	Hull Platforms and Flats
141	1st Platform
142	2nd Platform
143	3rd Platform
144	4th Platform
145	5th Platform
149	Flats
150	Deck House Structure
151	Deckhouse Structure to First Level
152	1st Deckhouse Level
153	2nd Deckhouse Level
154	3rd Deckhouse Level
155	4th Deckhouse Level
156	5th Deckhouse Level
157	6th Deckhouse Level
158	7th Deckhouse Level
159	8th Deckhouse Level and Above
160	Special Structures
161	Structural Castings, Forgings, and Equiv. Weldments
162	Stacks and Macks (Combined Stack and Mast)
163	Sea Chests
164	Ballistic Plating

SHIP WORK BREAKDOWN STRUCTURE TITLE

165	Sonar Domes
166	Sponsons
167	Hull Structural Closures
168	Deckhouse Structural Closures
169	Special Purpose Closures and Structures
170	Masts, Kingposts, and Service platforms
171	Masts, Towers, Tetrapods
172	Kingposts and Support Frames
179	Service Platforms
180	Foundations
181	Hull Structure Foundations
182	Propulsion Plant Foundations
183	Electric Plant Foundations
184	Command and Surveillance Foundations
185	Auxiliary Systems Foundations
186	Outfit and Furnishings Foundations
187	Armament Foundations
190	Special Purpose Systems
191	Ballast, Fixed or Fluid, and Buoyancy Units
192	Compartment Testing
195	Erection of Sub Sections (Progress Report Only)
198	Free Flooding Liquids
199	Hull Repair Parts and Special Tools

Group 2 Propulsion Plant

200	Propulsion Plant, General
201	General Arrangement-Propulsion Drawings
202	Machinery Plant Central Control Systems
210	Energy Generating System (Nuclear)
211	(Reserved)
212	Nuclear Steam Generator
213	Reactors
214	Reactor Coolant System
215	Reactor Coolant Service System
216	Reactor Plant Auxiliary Systems
217	Nuclear Power Control and Instrumentation
218	Radiation Shielding (Primary)
219	Radiation Shielding (Secondary)
220	Energy Generating System (Non-Nuclear)
221	Propulsion Boilers
222	Gas Generators

SHIP WORK BREAKDOWN STRUCTURE TITLE

223	Main Propulsion Batteries
224	Main Propulsion Fuel Cells
230	Propulsion Units
231	Propulsion Steam Turbines
232	Propulsion Steam Engines
233	Propulsion Internal Combustion Engines
234	Propulsion Gas Turbines
235	Electric Propulsion
236	Self-Contained Propulsion Systems
237	Auxiliary Propulsion Devices
238	Secondary Propulsion (Submarines)
239	Emergency Propulsion (Submarines)
240	Transmission and Propulsor Systems
241	Propulsion Reduction Gears
242	Propulsion Clutches and Couplings
243	Propulsion Shafting
244	Propulsion Shaft Bearings
245	Propulsory
246	Propulsor Shrouds and Ducts
247	Water Jet Propulsory
248	Lift System Fans and Ducting
250	Propulsion Support Sys. (Except Fuel and Lube Oil)
251	Combustion Air System
252	Propulsion Control System
253	Main Steam Piping System
254	Condensers and Air Ejectors
255	Feed and Condensate System
256	Circulating and Cooling Sea Water System
258	H.P. Steam Drain System
259	Uptakes (Inner Casing)
260	Propulsion Support Systems (Fuel and Lube Oil)
261	Fuel Service System
262	Main Propulsion Lube Oil System
263	Shaft Lube Oil System (Submarines)
264	Lube Oil Fill, Transfer, and Purification
290	Special Purpose Systems
298	Propulsion Plant Operating Fluids
299	Propulsion Plant Repair Parts and Special Tools

Group 3 Electric Plant

300	Electric Plant, General
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SHIP WORK BREAKDOWN STRUCTURE TITLE

301	General Arrangement-Electrical Drawings
302	Motors and Associated Equipment
303	Protective Devices
304	Electric Cables
305	Electrical Designating and Marking
310	Electric Power Generation
311	Ship Service Power Generation
312	Emergency Generators
313	Batteries and Service Facilities
314	Power Conversion Equipment
320	Power Distribution Systems
321	Ship Service Power Cable
322	Emergency Power Cable System
323	Casualty Power Cable System
324	Switchgear and Panels
330	Lighting System
331	Lighting Distribution
332	Lighting Fixtures
340	Power Generation Support Systems
341	SSTG Lube Oil
342	Diesel Support Systems
343	Turbine Support Systems
390	Special Purpose Systems
398	Electric Plant Operating Fluids
399	Electric Plant Repair Parts and Special Tools

Group 4 Command and Surveillance

400	Command and Surveillance, General
401	General Arrangement-Command and Surveillance
402	Security Requirements
403	Personnel Safety
404	Radio Frequency Transmission Lines
405	Antenna Requirements
406	Grounding and Bonding
407	Electromagnetic Interference Reduction (EMI)
408	System Test Requirements
410	Command and Control Systems
411	Data Display Group
412	Data Processing Group
413	Digital Data Switchboards
414	Interface Equipment

SHIP WORK BREAKDOWN STRUCTURE TITLE

415	Digital Data Communications
417	Command and Control Analog Switchboards
420	Navigation Systems
421	Non-Electrical/Electronic Navigation Aids
422	Electrical Navigation Aids (Incl Navig. Lights)
423	Electronic Navigation Systems, Radio
424	Electronic Navigation Systems, Acoustical
425	Periscopes
426	Electrical Navigation Systems
427	Inertial Navigation Systems
428	Navigation Control Monitoring
430	Interior Communications
431	Switchboards for I.C. Systems
432	Telephone Systems
433	Announcing Systems
434	Entertainment and Training Systems
435	Voice Tubes and Message Passing Systems
436	Alarm, Safety, and Warning Systems
437	Indicating, Order, and Metering Systems
438	Integrated Control Systems
439	Recording and Television Systems
440	Exterior Communications
441	Radio Systems
442	Underwater Systems
443	Visual and Audible Systems
444	Telemetry Systems
445	TTY and Facsimile Systems
446	Security Equipment Systems
450	Surveillance Systems (Surface)
451	Surface Search Radar
452	Air Search Radar (2D)
453	Air Search Radar (3D)
454	Aircraft Control Approach Radar
455	Identification Systems (IFF)
456	Multiple Mode Radar
459	Space Vehicle Electronic Tracking
460	Surveillance Systems (Underwater)
461	Active Sonar
462	Passive Sonar
463	Multiple Mode Sonar
464	Classification Sonar
465	Bathythermograph
470	Countermeasures

SHIP WORK BREAKDOWN STRUCTURE TITLE

471	Active ECM (Incl Combination Active/Passive)
472	Passive ECM
473	Torpedo Decoys
474	Decoys (Other)
475	Degaussing
476	Mine Countermeasures
480	Fire Control Systems
481	Gun Fire Control Systems
482	Missile Fire Control Systems
483	Underwater Fire Control Systems
484	Integrated Fire Control Systems
489	Weapon Systems Switchboards
490	Special Purpose Systems
491	Electronic Test, Checkout, and Monitoring Equipment
492	Flight Control and Instrument Landing Systems
493	Non Combat Data Processing Systems
494	Meteorological Systems
495	Special Purpose Intelligence Systems
498	Command and Surveillance Operating Fluids
499	Command and Surv. Repair Parts and Special Tools

Group 5 Auxiliary Systems

500	Auxiliary Systems, General
501	General Arrangement-Auxiliary Systems Drawings
502	Auxiliary Machinery
503	Pumps
504	Instrument and Instrument Boards
505	General Piping Requirements
506	Overflows, Air Escapes, and Sounding Tubes
507	Machinery and Piping Designation and Marking
508	Thermal Insulation for Piping and Machinery
509	Thermal Insulation for Vent and A/C Ducts
510	Climate Control
511	Compartment Heating System
512	Ventilation System
513	Machinery Space Ventilation System
514	Air Conditioning System
515	Air Revitalization Systems (Submarines)
516	Refrigeration System
517	Auxiliary Boilers and Other Heat Sources
520	Sea Water Systems

SHIP WORK BREAKDOWN STRUCTURE TITLE

521	Firemain and Flushing (Sea Water) System
522	Sprinkler System
523	Washdown System
524	Auxiliary Sea Water System
526	Scuppers and Deck Drains
527	Firemain Actuated Services-Other
528	Plumbing Drainage
529	Drainage and Ballasting System
530	Fresh Water Systems
531	Distilling Plant
532	Cooling Water
533	Potable Water
534	Aux. Steam and Drains Within Machinery Box
535	Aux. Steam and Drains Outside Machinery Box
536	Auxiliary Fresh Water Cooling
540	Fuels and Lubricants, Handling and Storage
541	Ship Fuel and Fuel Compensating System
542	Aviation and General Purpose Fuels
543	Aviation and General Purpose Lubricating Oil
544	Liquid Cargo
545	Tank Heating
549	Special Fuel and Lubricants, Handling and Stowage
550	Air, Gas, and Misc. Fluid Systems
551	Compressed Air Systems
552	Compressed Gases
553	O ₂ N ₂ System
554	LP Blow
555	Fire Extinguishing Systems
556	Hydraulic Fluid System
557	Liquid Gases, Cargo
558	Special Piping Systems.
560	Ship Control Systems
561	Steering and Diving Control Systems
562	Rudder
563	Hovering and Depth Control (Submarines)
564	Trim System (Submarines)
565	Trim and Heel Systems (Surface Ships)
566	Diving Planes and Stabilizing Fins (Submarines)
567	Strut and Foil Systems
568	Maneuvering Systems
570	Underway Replenishment Systems
571	Replenishment-At-Sea Systems
572	Ship Stores and Equip. Handling Systems

SHIP WORK BREAKDOWN STRUCTURE TITLE

573	Cargo Handling Systems
574	Vertical Replenishment Systems
580	Mechanical Handling Systems
581	Anchor Handling and Stowage Systems
582	Mooring and Towing Systems
583	Boats, Boat Handling and Stowage Systems
584	Mechanically Operated Door, Gate, Ramp, Turntable Sys.
585	Elevating and Retracting Gear
586	Aircraft Recovery Support Systems
587	Aircraft Launch Support Systems
588	Aircraft Handling, Servicing and Stowage
589	Miscellaneous Mechanical Handling Systems
590	Special Purpose Systems
591	Scientific and Ocean Engineering Systems
592	Swimmer and Diver Support and Protection Systems
593	Environmental Pollution Control Systems
594	Submarine Rescue, Salvage, and Survival Systems
595	Towing, Launching and Handling for Underwater Sys.
596	Handling Sys. for Diver and Submersible Vehicles
597	Salvage Support Systems
598	Auxiliary Systems Operating Fluids
599	Auxiliary Systems Repair Parts and Tools

Group 6 Outfit and Furnishings

600	Outfit and Furnishings, General
601	General Arrangement-Outfit and Furn. Drawings
602	Hull Designating and Marking
603	Draft Marks
604	Locks, Keys, and Tags
605	Rodent and Vermin Proofing
610	Ship Fittings
611	Hull Fittings
612	Rails, Stanchions, and Lifelines
613	Rigging and Canvas
620	Hull Compartmentation
621	Non-Structural Bulkhead
622	Floor Plates and Gratings
623	Ladders
624	Non-Structural Closures
625	Airports, Fixed Portlights, and Windows
630	Preservatives and Coverings

SHIP WORK BREAKDOWN STRUCTURE TITLE

631	Painting
632	Zinc Coating
633	Cathodic Protection
634	Deck Covering
635	Hull Insulation
636	Hull Damping
637	Sheathing
638	Refrigerated Spaces
639	Radiation Shielding
640	Living Spaces
641	Officer Berthing and Messing Spaces
642	Noncommissioned Officer Berthing and Messing Spaces
643	Enlisted Personnel Berthing and Messing Spaces
644	Sanitary Spaces and Fixtures
645	Leisure and Community Spaces
650	Service Spaces
651	Commissary Spaces
652	Medical Spaces
653	Dental Spaces
654	Utility Spaces
655	Laundry Spaces
656	Trash Disposal Spaces
660	Working Spaces
661	Offices
662	Machinery Control Centers Furnishings
663	Electronics Control Centers Furnishings
664	Damage Control Stations
665	Workshops, Labs, Test Areas (Incl Portable Tools, Equip)
670	Stowage Spaces
671	Lockers and Special Stowage
672	Storerooms and Issue Rooms
673	Cargo Stowage
690	Special Purpose Systems
698	Outfit and Furnishings Operating Fluids
699	Outfit and Furnish. Repair Parts and Special Tools

Group 7 Armament

700	Armament, General
701	General Arrangement-Weaponry Systems
702	Armament Installations
703	Weapons Handling and Stowage, General

SHIP WORK BREAKDOWN STRUCTURE TITLE

710	Guns and Ammunition
711	Guns
712	Ammunition Handling
713	Ammunition Stowage
720	Missiles and Rockets
721	Launching Devices (Missiles and Rockets)
722	Missile, Rocket, and Guidance Capsule Handling Sys.
723	Missile and Rocket Stowage
724	Missile Hydraulics
725	Missile Gas
726	Missile Compensating
727	Missile Launcher Control
728	Missile Heating, Cooling, Temperature Control
729	Missile Monitoring, Test and Alignment
730	Mines
731	Mine Launching Devices
732	Mine Handling
733	Mine Stowage
740	Depth Charges
741	Depth Charge Launching Devices
742	Depth Charge Handling
743	Depth Charge Stowage
750	Torpedoes
751	Torpedo Tubes
752	Torpedo Handling
753	Torpedo Stowage
754	Submarine Torpedo Ejection
760	Small Arms and Pyrotechnics
761	Small Arms and Pyrotechnic Launching Devices
762	Small Arms and Pyrotechnic Handling
763	Small Arms and Pyrotechnic Stowage
770	Cargo Munitions
772	Cargo Munitions Handling
773	Cargo Munitions Stowage
780	Aircraft Related Weapons
782	Aircraft Related Weapons Handling
783	Aircraft Related Weapons Stowage
790	Special Purpose Systems
792	Special Weapons Handling
793	Special Weapons Stowage
797	Misc. Ordnance Spaces
798	Armament Operating Fluids
799	Armament Repair Parts and Special Tools

SHIP WORK BREAKDOWN STRUCTURE TITLE

Group 8 integration/Engineering (Shipbuilder Response)

800	Integration/Engineering (Shipbuilder Response)
801	Shipbuilders Information Drawings
802	Contract Drawings
803	Standard Drawings
804	Type Drawings
806	Study Drawings
810	Production Engineering
811	Configuration Management
812	Change Proposals, Scoping and Shipchecking
813	Planning and Production Control
820	Special Drawings for Nuclear Propulsion Systems
830	Design Support
831	Construction Drawings
832	Specifications
833	Weight Engineering
834	Computer Programs
835	Engineering Calculations
836	Models and Mockups
837	Photographs
838	Design/Engineering Liaison
839	Lofting
840	Quality Assurance
841	Tests and Inspection, Criteria, and Procedures
842	Trials Agenda Preparation, Data Collection and Anal.
843	Inclining Experiment and Trim Dive
844	Combat Systems Checkout Criteria and Procedures
845	Certification Standards
850	Integrated Logistic Support Engineering
851	Maintenance
852	Support and Test Equipment
853	supply support
854	Transportation
855	Engineering Drawings and Specifications
856	Technical Manuals and Other Data
857	Facilities
858	Personnel and Training
859	Training Equipment
890	Special Purpose Items
891	Safety

SHIP WORK BREAKDOWN STRUCTURE TITLE

892	Human Factors
893	Standardization
894	Value Engineering
895	Reliability and Maintainability
896	Data Management
897	Project Management

Group 9 Ship Assembly and Support Services

900	Ship Assembly and Support Services
901	901 Thru 979 Reserved for Ident. of Assemblies
980	Contractual and Production Support Service
981	Insurance
982	Trials
983	Delivery
984	Open and Inspect (Conversions Only)
985	Fire and Flooding Protection
986	Tests and Inspection
987	Weighing and Recording
988	Contract Data Requirements (Administration)
989	Fitting-Out
990	Construction Support
991	Staging, Scaffolding, and Cribbing
992	Temporary Utilities and Services
993	Material Handling and Removal
994	Cleaning Services
995	Molds and Templates, Jigs, Fixtures, and Spec. Tools
996	Launching
997	Drydocking